TYREX

Deployment Guide

On-Premises

Document Versions

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Summary

This document explains how to deploy the On-premises version of the TYREX solution.

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1. INTRODUCTION

The TYREX solution can be quickly and easily deployed by your staff on your premises in various architecture types to meet your needs. When deployed in network infrastructure, TYREX complies with strict company security policies by storing all data locally.

This guide will help you set up your TYREX technology.

Note: A glossary is provided to explain commands and acronyms.

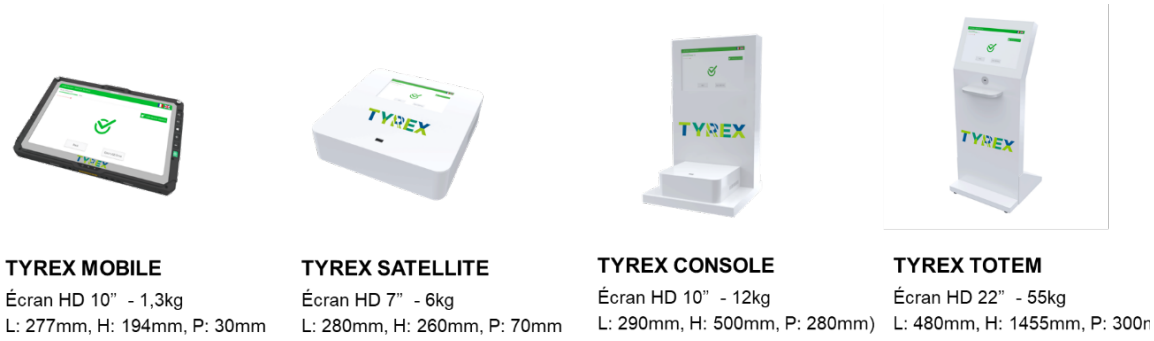

Figure 1: Our TYREX devices.

2. ARCHITECTURE

The On-premises architecture of the TYREX solution requires the management console to be installed in your infrastructure.

Management server updates are sent by TYREX from our secure central update infrastructure. This type of solution requires your management server to maintain a secure connection to our central update management platform.

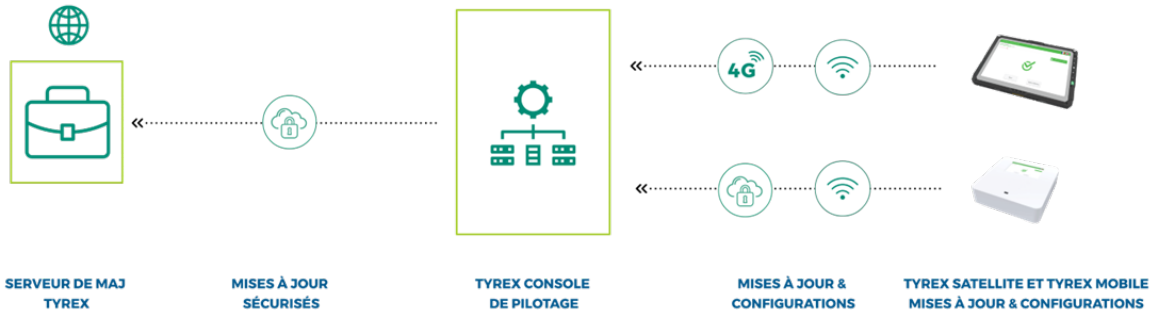

Figure 2: General On-premises architecture

3. REQUIREMENTS

3.1. HARDWARE

TYREX provides a VM to be deployed in your virtualization environment, called the management server. This management server is managed through the management console.

To deploy this, you need:

- a virtualization host compatible with OVF/VMDK (ESXI), VHD (Hyper-V), or KVM (Proxmox);
- At least **500 GBs** of disk space, **8 GBs of RAM**, and **4 vCPUs**.

3.2. NETWORK

The management server must have internet access either directly or through an HTTP proxy. SOCKS or TCP proxies are not currently supported.

> **Warning**: Please note that our server update system uses client-side SSL certificates to authenticate with our central server. To do this through a proxy, you will need to use the CONNECT method. Some proxy software does not support CONNECT calls with authentication through login/password, so you will need to use another authentication method such as an IP or MAC filter.

This diagram shows the flows used to run your TYREX solution:

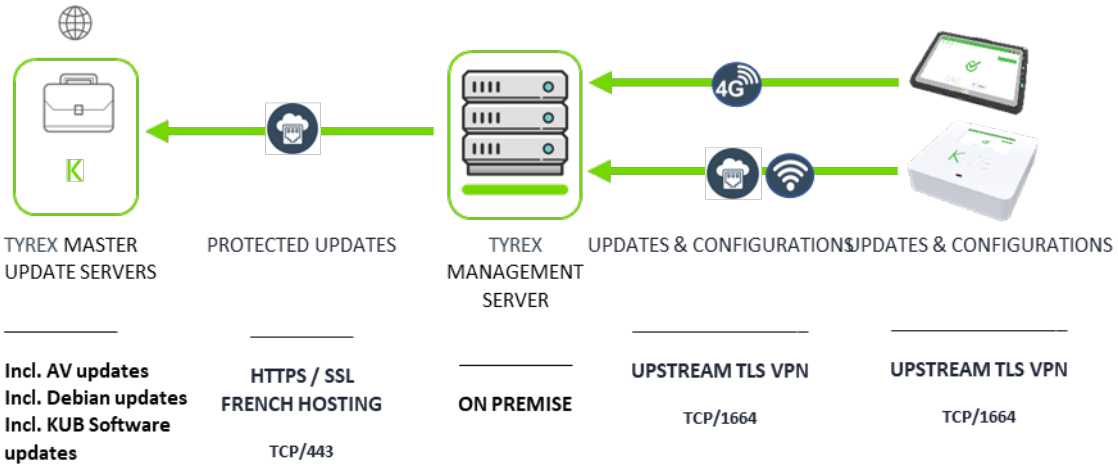

Figure 3: On-premises flow chart

Port routing

Sources	Destination	Protocol	Port	Verification	Notes
Management server	Puppet.kub-cleaner.tech Apt.kub-cleaner.tech	TCP	443, DNS	Telnet puppet.kub-cleaner.tech 443 Telnet Apt.kub-cleaner.tech 443	Centralizes antivirus and system updates to distribute them to TYREX devices.
Admin network	Management server	TCP	9122, 80, and/or 443	Run this command: netstat — tlpn l grep [Port] to check the ports are open	SSH access to the management server (optional) HTTP or HTTPS access to the management console
TYREX device	Management server	TCP	1664	NA	Each device deployed in the IS uses OpenVPN to communicate with the management server
TYREX device	NFS uploading	UDP/TCP	2049	NA	Allows the TYREX device to send clean files to an NFS upload
TYREX device	SAMBA uploads	TCP	139/445	NA	Allows the TYREX station to send clean files to a SAMBA share
TYREX device	SFTP sharing	TCP	22/9122	NA	Allows the TYREX device to send clean files to an SFTP upload
TYREX server	SIEM	UDP/TCP	*	NA	Allows logs from the solution to be sent to a log collector for analysis

Table 1: On-premises flow matrix

4. Installing the Management Server

4.1. Introduction

Before beginning, you first need to deploy the VM provided by TYREX into your virtualization environment.

Please refer to your virtualization solution's manual for this step. Once the application has been imported, check that the virtual machine's settings meet our requirements.

4.2. Authentication

The default **root** password for the virtual machine is "**CyberKUBwall1720**". The default username, without any privileges, is "**netlor**". This user can be used as desired and, specifically, is the only one able to connect via SSH.

Figure 4: Authentication command lines.

4.3. Network Configuration

To enable the management server to communicate with our infrastructure, you first need to modify its network configuration to use your own address.

To change the network configuration, modify the file located at /etc/network/interfaces. This file is a standard Debian network interfaces config file.

Example: You want to configure a static IP on my virtual machine. You are going to use the following commands to set the new IP address, then restart the services.

Commands to run on your virtual machine's Linux console.

Edit the network interface file:

```
nano /etc/network/interfaces
```

↳ Define the new IP address for my infrastructure:

```
auto eth0
iface eth0 inet static
address 192.168.144.10
gateway 10.102.152.254
netmask 255.255.255.0
dns-nameservers 8.8.8.8
```

```
  GNU nano 3.2                          /etc/network/interfaces

# This file describes the network interfaces available on your system
# and how to activate them. For more information, see interfaces(5).

source /etc/network/interfaces.d/*

# The loopback network interface
auto lo
iface lo inet loopback

# The primary network interface
auto ens192
iface ens192 inet static
        address 10.103.3.1/24
        gateway 10.103.3.254
        # dns-* options are implemented by the resolvconf package, if installed
        dns-nameservers 8.8.8.8
```

Figure 5: Command lines for defining IP address

↳ Then relaunch the network service to apply your new IP address to your network card:

```
/etc/init.d/networking restart
```

Now check that feeds are open and that they allow the server to access TYREX infrastructure:

To do this, run the necessary network tests:

```
wget apt.kub-cleaner.tech
```

```
root@kubserver:~# wget apt.kub-cleaner.tech
--2023-11-22 10:52:11--  http://apt.kub-cleaner.tech/
Résolution de apt.kub-cleaner.tech (apt.kub-cleaner.tech)… 185.37.30.72
Connexion à apt.kub-cleaner.tech (apt.kub-cleaner.tech)|185.37.30.72|:80… connecté.
requête HTTP transmise, en attente de la réponse… 302 Found
Emplacement : https://puppet.kub-cleaner.tech/ [suivant]
--2023-11-22 10:52:11--  https://puppet.kub-cleaner.tech/
Résolution de puppet.kub-cleaner.tech (puppet.kub-cleaner.tech)… 185.37.30.72
Connexion à puppet.kub-cleaner.tech (puppet.kub-cleaner.tech)|185.37.30.72|:443… connecté.
Erreur : le certificat de « puppet.kub-cleaner.tech » n'est pas de confiance.
Erreur: The certificate of « puppet.kub-cleaner.tech » doesn't have a known issuer.
```

```
root@kubserver:~# wget apt.kub-cleaner.tech
--2023-12-20 12:15:35--  http://apt.kub-cleaner.tech/
Résolution de apt.kub-cleaner.tech (apt.kub-cleaner.tech)… 185.37.30.72
Connexion à apt.kub-cleaner.tech (apt.kub-cleaner.tech)|185.37.30.72|:80… connecté.
requête HTTP transmise, en attente de la réponse… 302 Found
Emplacement : https://puppet.kub-cleaner.tech/ [suivant]
--2023-12-20 12:15:35--  https://puppet.kub-cleaner.tech/
Résolution de puppet.kub-cleaner.tech (puppet.kub-cleaner.tech)… 185.37.30.72
Connexion à puppet.kub-cleaner.tech (puppet.kub-cleaner.tech)|185.37.30.72|:443… connecté.
Erreur : le certificat de « puppet.kub-cleaner.tech » n'est pas de confiance.
Erreur: The certificate of « puppet.kub-cleaner.tech » doesn't have a known issuer.
root@kubserver:~#
```

Figure 6: Test command lines

- telnet puppet.kub-cleaner.tech 443

```
root@kubserver:~# telnet puppet.kub-cleaner.tech 443
Trying 185.37.30.72...
Connected to puppet.kub-cleaner.tech.
Escape character is '^]'.
```

```
root@kubserver:~# telnet puppet.kub-cleaner.tech 443
Trying 185.37.30.72...
Connected to puppet.kub-cleaner.tech.
Escape character is '^]'.
```

Figure 7: Test command lines

If your configuration is valid, run the command:

puppet agent -tv

```
root@kubserver:~# puppet agent -tv
Info: Using environment 'v2_0_x'
Info: Retrieving pluginfacts
Info: Retrieving plugin
Info: Loading facts
/opt/puppetlabs/puppet/cache/lib/puppet/type/gnupg_key.rb:91: warning: URI.escape is obsolete
```

```
/opt/puppetlabs/puppet/cache/lib/puppet/type/gnupg_key.rb:102: warning: URI.escape
is obsolete
/opt/puppetlabs/puppet/cache/lib/puppet/type/gnupg_key.rb:103: warning: URI.escape
is obsolete
/opt/puppetlabs/puppet/cache/lib/puppet/type/gnupg_key.rb:91:  warning:  URI.escape
is obsolete
/opt/puppetlabs/puppet/cache/lib/puppet/type/gnupg_key.rb:102: warning: URI.escape
is obsolete
/opt/puppetlabs/puppet/cache/lib/puppet/type/gnupg_key.rb:103: warning: URI.escape
is obsolete
Info: Caching catalog for <puppet cert name>
```

to run a manual update on the TYREX server and check that the server is able to update by connecting to our infrastructure.

```
root@kubserver:~# puppet agent -tv
Info: Using environment 'v2_0_x'
Info: Retrieving pluginfacts
Info: Retrieving plugin
Info: Loading facts
/opt/puppetlabs/puppet/cache/lib/puppet/type/gnupg_key.rb:91: warning: URI.escape is obsolete
/opt/puppetlabs/puppet/cache/lib/puppet/type/gnupg_key.rb:102: warning: URI.escape is obsolete
/opt/puppetlabs/puppet/cache/lib/puppet/type/gnupg_key.rb:103: warning: URI.escape is obsolete
/opt/puppetlabs/puppet/cache/lib/puppet/type/gnupg_key.rb:91: warning: URI.escape is obsolete
/opt/puppetlabs/puppet/cache/lib/puppet/type/gnupg_key.rb:102: warning: URI.escape is obsolete
/opt/puppetlabs/puppet/cache/lib/puppet/type/gnupg_key.rb:103: warning: URI.escape is obsolete
Info: Caching catalog for kub-srv-test-v2-01
```

Figure 8: Command lines to manually update the TYREX server

```
Info: /Stage[main]/Main/File[/srv/webconsole/migrate_notifications]: Scheduling refresh of Exec[webc
onsole_bootstrap]
Info: /Stage[main]/Main/File[/srv/webconsole/migrate_notifications]: Scheduling refresh of Exec[webc
onsole_bootstrap]
Notice: /Stage[main]/Main/File[/srv/webconsole/scripts/host/uuid_cleaner.sh]/owner: owner changed 'k
ubserver' to 'root' (corrective)
Notice: /Stage[main]/Main/File[/srv/webconsole/scripts/host/uuid_cleaner.sh]/group: group changed 'w
ww-data' to 'kubserver' (corrective)
Notice: /Stage[main]/Main/File[/srv/webconsole/scripts/host/PrebuildSufFiles_force.sh]/owner: owner
changed 'kubserver' to 'root' (corrective)
Notice: /Stage[main]/Main/File[/srv/webconsole/scripts/host/PrebuildSufFiles_force.sh]/group: group
changed 'www-data' to 'kubserver' (corrective)
Notice: /Stage[main]/Main/Exec[suf]/returns: executed successfully (corrective)
Notice: /Stage[main]/Main/Exec[suf]: Triggered 'refresh' from 1 event
Notice: /Stage[main]/Main/Mysql::Db[monitoring]/Mysql_database[monitoring]/charset: charset changed
'utf8mb3' to 'utf8' (corrective)
Notice: /Stage[main]/Main/Mysql::Db[monitoring]/Mysql_database[monitoring]/collate: collate changed
'utf8mb3_general_ci' to 'utf8_general_ci' (corrective)
Notice: /Stage[main]/Main/Mysql::Db[kub]/Mysql_database[kub]/charset: charset changed 'utf8mb3' to '
utf8' (corrective)
Notice: /Stage[main]/Main/Mysql::Db[kub]/Mysql_database[kub]/collate: collate changed 'utf8mb3_gener
al_ci' to 'utf8_general_ci' (corrective)
Notice: /Stage[main]/Main/File[/srv/webconsole/bootstrap.sh]/owner: owner changed 'kubserver' to 'ro
ot' (corrective)
Notice: /Stage[main]/Main/File[/srv/webconsole/bootstrap.sh]/group: group changed 'www-data' to 'kub
server' (corrective)
Info: /Stage[main]/Main/File[/srv/webconsole/bootstrap.sh]: Scheduling refresh of Exec[webconsole_bo
otstrap]
Info: /Stage[main]/Main/File[/srv/webconsole/bootstrap.sh]: Scheduling refresh of Exec[webconsole_bo
otstrap]
Notice: /Stage[main]/Main/Exec[webconsole_bootstrap]: Triggered 'refresh' from 7 events
Notice: /Stage[main]/Main/Service[WebConsole]/ensure: ensure changed 'stopped' to 'running' (correct
ive)
Info: /Stage[main]/Main/Service[WebConsole]: Unscheduling refresh on Service[WebConsole]
Notice: Applied catalog in 1717.80 seconds
root@kubserver:~#
```

Figure 9: Command lines to manually update the TYREX server

Warning: If you see this appear, the update process is already in progress. If you are in the initial installation phase, we recommend deleting the agent_catalog.lock file and re-running the command.

```
root@kubserver:~# puppet agent -tv
Notice: Run of Puppet configuration client already in progress; skipping  (/opt/puppetlabs/puppet/ca
che/state/agent_catalog_run.lock exists)
root@kubserver:~#
```

Figure 10: Command lines to manually update the TYREX server

Access the management console using the following link in a browser: http://server.ip.address.or.fqdn/.

Note: If you need to add a proxy to access the server through SSH, or to add certificates, go to the **Advanced Configuration** section below.

4.4. Logging in to the Management Console for the first time

The default login details to access the platform are:

- Username: **admin**
- Password: **admin**

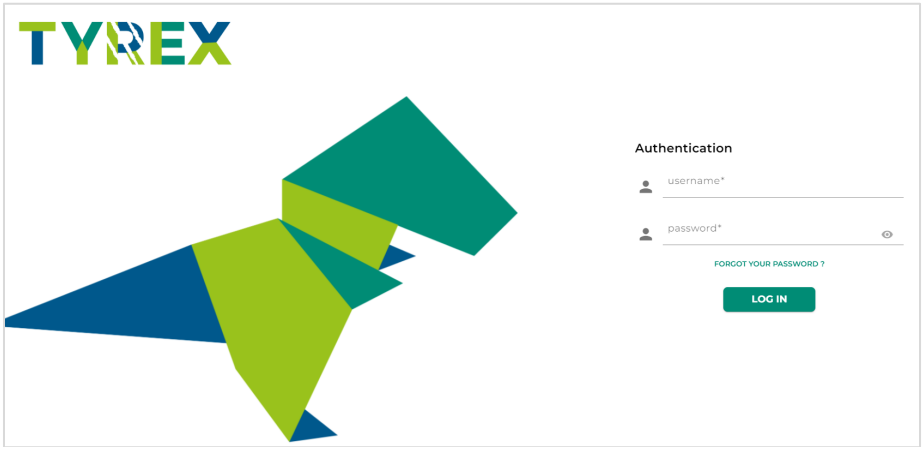

Figure 11: Management console authentication screen.

Note: When logging in for the first time, we recommend changing this password by clicking on your profile name in the bottom left of the screen and selecting the Profile option.

On that page, you can then change your name, email address, and password.

Warning: As part of the On-premises mode, we recommend changing the server's PGP key before configuring the TYREX devices. Go to the following section for instructions on how to do this: 2.9.1.2 of the On-Premises Administrator Guide.

5. COMMISSIONING

5.1. CREATING ACCOUNTS AND GROUPS

- Log in to the management console (server IP address).

- Go to the **Platform** section and select **Users**.

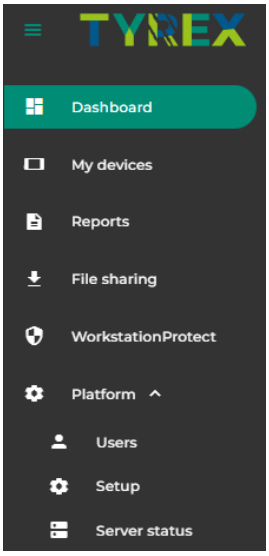

Figure 12: Users section

- Click the

button to add a **user**.

- Fill out the form with the required information.

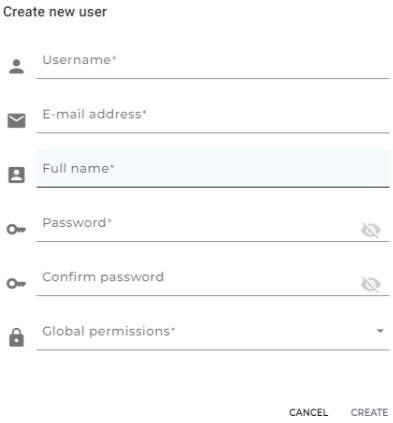

Figure 13: User creation form

Warning: Passwords must have a minimum of 10 characters including at least: 1 capital letter, 1 number, and 2 special characters (!@#$ %^&*).

- Click the **Groups** tab, then the + create group button to create a group.
- Enter the desired **Group name**.

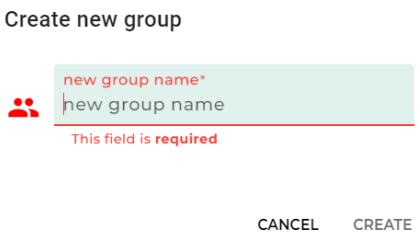

Figure 14: Creating a group

- Once the group is created, click the **Edit group access** icon in the **Actions** column to add users to this group.
- Add all users who will have access to TYREX devices to this group.

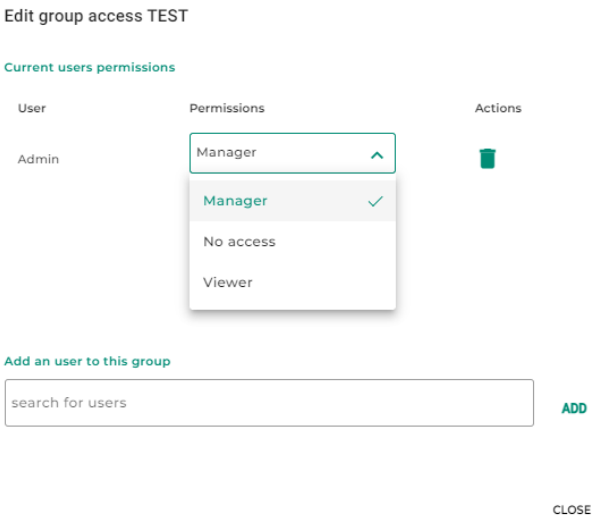

Figure 15: Adding permissions

Warning: Take care with user permissions.

5.2. Creating a TYREX Device

- Go to the **My devices** section, then click the + button to add a TYREX device.
- Fill out the form. The **Serial number** is on a label on the underside of the TYREX device.

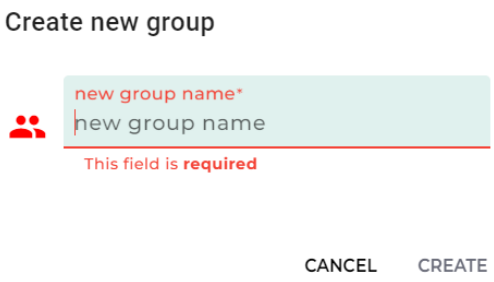

Figure 16: Adding a TYREX device

5.3. Adding Antivirus Engines

- To modify the active antivirus, select your device in the **My devices** section.
- Click the 3 vertical dots in the top right of your screen, then **Edit enabled engines**.

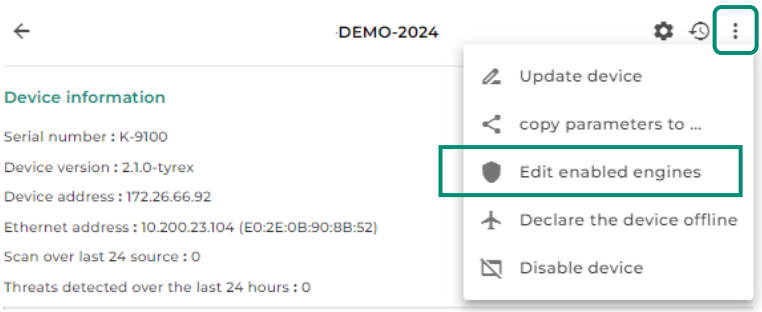

Figure 17: Antivirus management

- Select the **antivirus** you want to use.

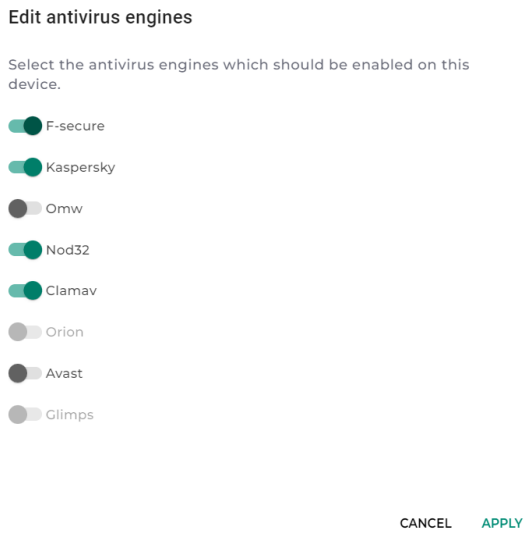

Figure 18: List of active antivirus/antimalware

5.4. Associating with the Server

- Select your TYREX device in the **My devices** section, then click **Offline file/Associate with server.**

- Fill out the form with your server's IP or FQDN address and the **Proxy configuration** if needed. Click **Generate** to create an update file (ABAV_UPD.SIG).

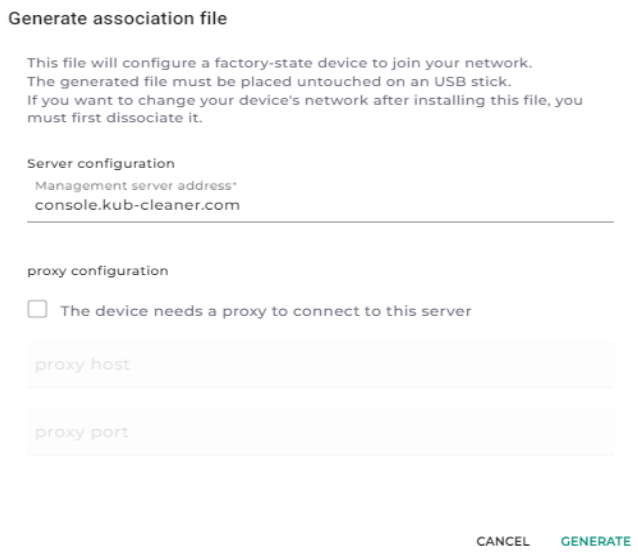

Figure 19: Generating a proxy configuration

- Put this file in the root directory of a USB device and insert the device into the TYREX device. Click this file in the popup which appears.

- The TYREX device will restart.

> **Note**: We recommend using an FQDN to link TYREX devices. When changing the server's IP address, you must also unlink each device configured with the old IP address, then relink it with the new IP address, which can be a lengthy process.

5.5. Network Configuration

> **Note**: If a device is offline, this network section cannot be configured.

- Once the TYREX device has been restarted, go to the **My devices** section, then click **Offline file/Network configuration**.

- Fill out the form with your network's details.

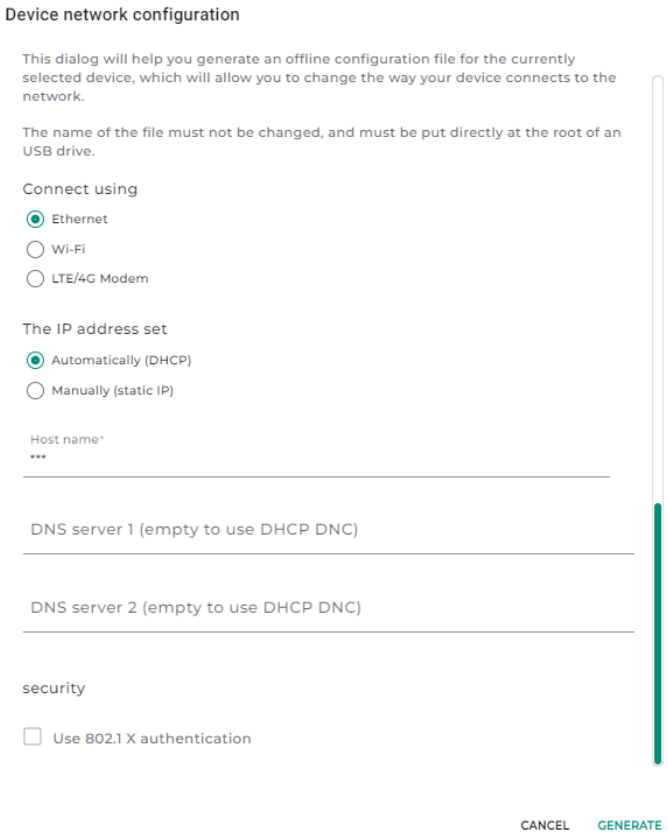

Figure 20: Device network configuration.

- Click **Generate**, then follow the same steps described in the Associating with the Server section, using the "ABAV_UPD.sig" file.

5.6. FINALIZATION

- Allow the TYREX device to connect and update its antivirus and system.

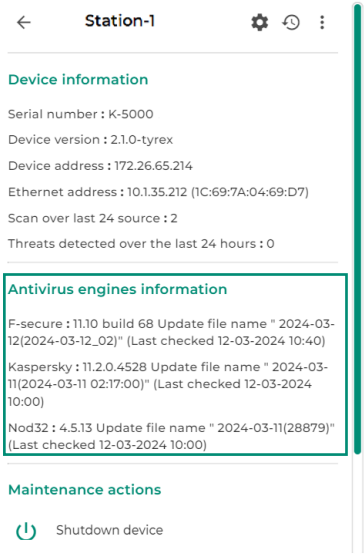

Figure 21: Defining and updating the antivirus

- Once the TYREX device is connected and the antivirus is up to date, the TYREX device is ready to use.

6. Advanced Configuration

6.1. Accessing the Management Console over Internet via Proxy

- If your server requires an HTTP proxy to access the internet, you can execute script/root/Tools/SetProxy.sh in the root. You must also enter the proxy IP address and the port (and the username/password, if required) in the following format: **./SetProxy.sh IP_ADDRESS PORT [LOGIN] [PASSWORD]**
  - The "Login" and "Password" parameters are between brackets, as they are optional. You must enter these values without brackets.

```
kubserver:~/Tools# ./SetProxy.sh 10.101.115.249 8080 Username Password
cp: cannot stat '/etc/apt/apt.conf.d/01proxy': No such file or directory

The system proxy has been set to 10.101.115.249:8080.

kubserver:~/Tools# _
```

Figure 22: Command lines for accessing the management console

- Once this script is executed, you must restart your server to apply the modification to all services.
- Run a reboot command:

reboot

- You can use the UnsetProxy.sh script to delete the proxy.

Note: Our server update system (based on Puppet) uses client-side SSL certificates to authenticate with our central server. To do this through a proxy, you will need to use the CONNECT method.

Some proxy software does not support CONNECT calls with authentication through login/password, so you will need to use another authentication method such as an IP or MAC filter.

6.2. SSH Access

An OpenSSH server is enabled on port 9122, which allows you to connect to the server remotely using the “netlor” user on the server.

For security reasons, we force authentication via a public key, and deactivate authentication by password.

To log in, you must register your public key in the file **/home/netlor/. ssh/authorized_keys**. By default, there is no key registered, so no one will be able to log in.

Note: Authentication by public key is forced, which means that the OpenSSH configuration is overwritten each time the platform is updated.

You have the **SSH Access Guide** available for guidance if needed.

6.3. Configuring HTTPS (SSL) for the Management Console

- To enable HTTPS (SSL) on the management console, you must put:
  - the certificate (crt) in: **/etc/nginx/ssl/webconsole.crt**
  - the private key (key) in: **/etc/nginx/ssl/webconsole.key**

- and modify (or create) the file /etc/kubstate/facts.conf to include the following lines:

```
webconsole_enable_ssl=true
webconsole_disable_http=true
```

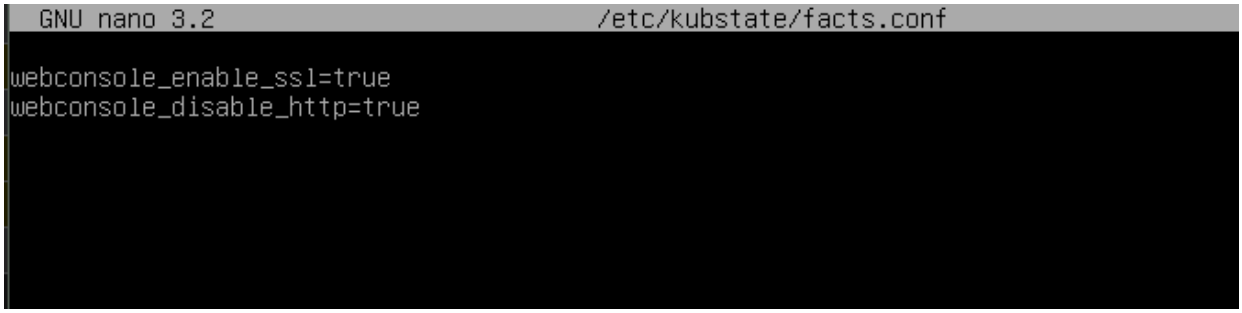

Figure 23: Command lines to enable SSL

- The first line enables SSL support, while the second disables classic HTTP access.
- After making these changes, you must force a system update (enter the command “puppet agent -tv”, then press **Enter** to confirm) to update the config files with these changes.
- You can redefine the values to “false” to cancel this change and return to HTTP access.

6.4. Configuring the Network Interface to Use the Management Console

- Your configuration may require use of a physical network interface that is different to the interface used to access the internet. While we try to detect the interface automatically, it is possible that the web interface will listen to the wrong network interface. To force the correct network interface, create (or edit) the file **/etc/kubstate/facts.conf** and add the following line: **webconsole_eth_if=ethX**

- Replace "ethX" with the name of the interface to use for the management console (such as "eth1").

```
  GNU nano 3.2                    /etc/kubstate/facts.conf

webconsole_eth_if=ethX
```

Figure 24: Lines to configure the network interface

- After making this change, force an update (enter the "puppet agent -tv" command and press **Enter** to confirm) to update the configuration with these changes.
- Command: puppet agent -tv

6.5. Disabling Automatic Updates

Warning: If you disable automatic updates, this will disable all remote updates, including management console updates, security system updates, TYREX updates, antivirus updates, and platform updates.

- If you want to disable automatic updates, you can stop the Puppet agent by running the following command:

systemctl stop puppet systemctl disable puppet

```
kubserver:~/Tools# systemctl stop puppet
kubserver:~/Tools#
```

Figure 25: Command lines to disable updates

```
kubserver:~/Tools# systemctl disable puppet
Removed /etc/systemd/system/multi-user.target.wants/puppet.service.
kubserver:~/Tools#
```

Figure 26: Command lines to disable updates

- You can re-enable them using the following command:

systemctl start puppet systemctl enable puppet

```
kubserver:~/Tools# systemctl start puppet
kubserver:~/Tools#
```

Figure 27: Command lines to re-enable updates

```
kubserver:~/Tools# systemctl enable puppet
Created symlink /etc/systemd/system/multi-user.target.wants/puppet.service → /lib/systemd/system/pup
pet.service.
kubserver:~/Tools#
```

Figure 28: Command lines to re-enable updates

For more information, please see the **On-Premises Administrator Guide.**

7. GLOSSARY

This section defines certain terms to help you understand this guide.

Terminology for TYREX system applications

Term	Definition
Management server	Management server refers to the system part of the management server.
Management console	The management console is the web interface which allows you to administer the management server.
WorkstationProtect Agent	The WorkstationProtect agent is an option which allows TYREX devices to certify removable storage devices after scanning and decontamination. Installed on IT and OT workstations, this agent is used to block devices and/or files which have not been certified by a TYREX device.
TYREX device	TYREX devices are sheep-dips in the form of CONSOLES, TOTEMS, SATELLITES, or MOBILES which are used to scan USB devices.

Table 2: TYREX solution glossary

Linux Commands

Pwd ..... : allows you to find the path of the working directory (folder) you are currently in

cd ..... : allows you to navigate Linux files and directories

ls..... : allows you to display the contents of directory

cat ..... : allows you to list the contents of a file on the standard output

cp ..... : allows you to copy the files in the current directory to another directory

mv ..... : allows you to move files and rename files

mkdir ..... : allows you to create a new directory

rm ..... : allows you to delete directories and their contents

touch ..... : allows you to create a new file through the Linux command line

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